ADITYA UNIVERSITY

002164

PART-I

S.No: B304089

Programe	:	B.Tech
Hall Ticket No	:	24B11AI238
Name	:	Mamidala Jyothi Sivani
Examination	:	III SEMESTER END EXAMINATIONS REGULAR
Month-Year	:	Nov-25
Branch	:	Artificial Intelligence & Machine Learning
Course Code	:	241MA009
Course Name	:	Probability & Statistics
Date of Exam	:	18.11.2025

J. Pavan.

Signature of the Controller of Examination

M. Jyothi Sivani 18.11.25

Signature of the Student with date

18/11/25

Signature of the Invigilator with date

Serial No. Last Page Written	30

INSTRUCTIONS TO STUDENTS

1. The Answer Booklet contains 36 pages Ensure all the 36 pages are in proper order.
2. Student must verify the details of particulars in the PART - 1 i.e Name, Hall Ticket No., Examination, Course Name etc.,
3. In case of any deviation in the above or if the PART-1 is torn / damaged, report to the invigilator and return the damaged booklet.
4. You are prohibited from writing on or tampering the PART-1 except affixing the signature and serial number of last page written in the space provided.
5. Student is prohibited from:
   i. Writing their Hall Ticket No. and name in any part of the answer booklet.
   ii. Addressing the examiner in any manner whatsoever in the answer booklet. If they do so, their script will not be valued.
   iii. Writing religious symbols.
   iv. Either seeking or providing any assistance to the fellow students in the examinations.
   v. Possessing a manuscript or a printed matter, in any form, in the examination hall.
   vi. Bringing Mobile Phones / Cameras/ Bluetooth Devices / Programmable calculators etc.
   vii. Violation of the above mentioned instructions will be viewed as a case of malpractice, which is punishable offence.
6. Before beginning to answer any question, the students should write the correct number of that question in the margin provided. Answers written at different places for the same question will not be valued.
7. Students should write the answers, within the margins provided on both sides of the paper and on all the lines of each page. It is not necessary to begin each answer in a fresh page. Answers must be legibly written with blue or black pen.
8. Do not write anything except Question Number in the margin.
9. No loose sheets of papers will be allowed in the examination hall. No paper must be detached from or attached to the answer booklet except graph sheet.
10. Strike off all unused pages.
11. **NO ADDITIONAL ANSWER BOOKLETS/SHEETS WILL BE SUPPLIED.**

SPACE FOR ROUGH WORK

Q.No.

1 a) Given,

A, and B are the bags containing white, red, green balls.

Bag A contains : white = 1 $P(A) = \frac{1}{2}, P(B) = \frac{1}{2}$

Red = 2

green = 3 Total = 6

Bag B contains : white = 2

red = 3

green = 1 Total = 6

probability of getting a white ball from A = $\frac{1}{6}$

probability of getting a Red ball from A = $\frac{2}{6} = \frac{1}{3}$

probability of getting a green ball from A = $\frac{3}{6}$

Probability of getting a white ball from A = $\frac{2}{6}$

Probability of getting a Red ball from B = $\frac{3}{6}$

probability of getting a green ball from B = $\frac{1}{6}$

$\therefore P(W/A) = \frac{1}{6}, \quad P(R/A) = \frac{2}{6}, \quad P(g/A) = \frac{3}{6}$

$P(W/B) = \frac{2}{6}, \quad P(R/B) = \frac{3}{6} \quad P(g/B) = \frac{1}{6}$

$\therefore$ Two balls are drawn from a bag chocen at Random, those are white and Red.

probability of getting a white ball from Bag B

$P(B/W) = \quad P(B) \cdot P(B$

$$P\left(B/W\right) = \frac{P(B).\ P(W/B)}{P(A).P(W/A) + P(B).\ P(W/B)}$$

$$= \frac{\frac{1}{2}\cdot\frac{2}{6}}{\frac{1}{2}\cdot\frac{1}{6} + \frac{1}{2}\cdot\frac{2}{6}}$$

$$= \frac{1/6}{\frac{1}{12} + \frac{1}{6}}$$

$$= \frac{0.167}{0.25}$$

$$= 0.668 \quad \therefore 6\ \%$$

Probability of getting a Red ball from bag B.

$$P\left(B/R\right) = \frac{P(B).\ P(R/B)}{P(A).\ P(R/A) + P(B)\,P(R/B)}$$

$$= \frac{1/2 \cdot 3/6}{\frac{1}{2}\cdot\frac{2}{6} + \frac{1}{2}\cdot 3/6}$$

$$= \frac{3/12}{1/6 + 3/12}$$

$$= \frac{0.25}{0.4167} \qquad \therefore\ 6\ \%$$

$$= 0.59 \simeq 0.6$$

Final propabibity of Bag B for white & Red balls

$$= 0.668 + 0.59 = \cancel{1.258} \qquad \therefore 6\ \%$$

Q.No.

b) i, k.

sum of the probabilities $= 1$

$\Rightarrow 0 + k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$

$9k + 10k^2 = 1$

$10k^2 + 9k - 1 = 0$

$\Rightarrow k = 0.1, -1$

$\therefore k = 0.1 > 0$

ii, $P(X < 6)$

$= P(0) + P(1) + P(2) + P(3) + P(4) + P(5) + P(6)$

$= 0 + k + 2k + 2k + 3k + k^2 + 2k^2$

$= 8k + 3k^2$

$= 8(0.1) + 3(0.1)^2$

$= 0.8 + 0.03$

$= 0.83$

iii, $P(0 < X < 5)$

$= P(1) + P(2) + P(3) + P(4)$

$= k + 2k + 2k + 3k$

$= 8k$

$= 8(0.1)$

$= 0.8$

iv, mean

$\mu = \Sigma x_i p_i$

$= 0(0) + 1(k) + 2(2k) + 3(2k) + 4(3k) + 5(k^2)$

$+ 6(2k^2) + 7(7k^2 + k)$

$= 0 + k + 4k + 6k + 12k + 5k^2 + 12k^2 + 49k^2 + 7k$

$= 30k + 66k^2$

$= 30(0.1) + 66(0.1)^2$

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$= 3 + 66(0.01)$

$= 3 + 0.66$

$= 3.66$

(v) variance.

$\sigma^2 = \sum x_i^2 p_i - \mu^2$

$= 0^2(1) + 1^2(k) + 2^2(2k) + 3^2(2k) + 4^2(3k) + 5^2(k^2) + 6^2(2k^2) + 7^2(7k^2+k)$

$= 0 + k + 8k + 18k + 48k + 25k^2 + 72k^2 + 343k^2 + 49k$

$= 124k + 440k^2$

$= 124(0.1) + 440(0.01)$

$= 12.4 + 4.4$

$= 16.8$

2a) given,

$f(x) = k\, x^2 e^{-x}\,;\; x \geq 0$

$\int_{-\infty}^{\infty} f(x)\, dx = 1$

$\int_{-\infty}^{0} f(x)\, dx + \int_{0}^{\infty} f(x)\, dx = 1$

$0 + \int_{0}^{\infty} k.\, x^2 e^{-x} = 1$

$\int_{0}^{\infty} k.\, x^2 e^{-x} = 1$

$k\left[e^{-x}\left(\frac{x^2}{-1} - \frac{2x}{1} + \frac{2}{-1}\right)\right]_{0}^{\infty} = 1$

Q.No.

$$k\left[e^{-\infty}(0) - e^{0}\left(0-0+\frac{2}{-1}\right)\right] = 1$$

$$k\ (2) = 1$$

$$\therefore\ k = \frac{1}{2}.$$

i. mean

$$\mu = \int_0^\infty x\, f(x)\, dx$$

$$= \int_0^\infty x . k x^2 e^{-x}\, dx$$

$$= k \int_0^\infty x^3 e^{-x}\, dx$$

$$= \frac{1}{2}\left[e^{-x}\left(\frac{x^3}{-1} - \frac{3x^2}{1} + \frac{6x}{-1} - \frac{6}{1}\right)\right]_0^\infty$$

$$= \frac{1}{2}\left[e^{-\infty}(0) - e^{-0}(0-0+0-6)\right]$$

$$= \frac{1}{2}\left[0+6\right]$$

$$= 3$$

ii. Variance

$$\sigma^2 = \int_0^\infty x^2 f(x)\, dx - \mu^2$$

$$= \int_0^\infty x^2 . k . x^2 e^{-x}\, dx - \mu^2$$

$$= k \int_0^\infty x^4 e^{-x}\, dx - \mu^2$$

$$= \frac{1}{2} \int_0^\infty x^4 e^{-x}\, dx - 3^2$$

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$$= \frac{1}{2}\left[e^{-x}\left(\frac{x^4}{-1} - \frac{4x^3}{1} + \frac{12x^2}{-1} - \frac{24x}{1} + \frac{24}{-1} \right)\right]_0^{\infty} - \mu^2$$

$$= \frac{1}{2}\left[e^{-\infty}(0) - e^{0}(0-0+0-0-24)\right] - \mu^2$$

$$= \frac{1}{2}[24] - 3^2$$

$$= 12 - 9$$

$$= 4$$

b) given, $f(x) = \begin{cases} 0 & \text{if } x \le 1 \\ k(x-1)^3 & 1 \le x \le 3 \\ 1 & \text{if } x > 3 \end{cases}$

i, k.

$$\int_1^3 k(x-1)^3 = 1$$

$$\left[k \frac{(x-1)^4}{4} \right]_1^3 = 1$$

$$k\left[\frac{(3-1)^{4}}{4} - \frac{(1-1)^4}{4} \right] = 1$$

$$k\left[\frac{2^4}{4} - 0 \right] = 1$$

$$k\left[\frac{16}{4} \right] = 1$$

$$k = \frac{1}{4}$$

Q.No.

ii, p.d.f of x

$$= \int_1^3 \frac{1}{4}(x-1)^3\,dx$$

$$= \frac{1}{4}\left[\int_1^3 (x-1)^3\,dx\right]$$

$$= \frac{1}{4}\left[\left[\frac{(x-1)^4}{4}\right]_1^3\right]$$

$$= \frac{1}{4}\left[\frac{(3-1)^4}{4} - \frac{(1-1)^4}{4}\right]$$

$$= \frac{1}{4}\left[\frac{16}{4} - 0\right]$$

$$= 1/4\,[4]$$

$$= 1$$

iii, mean

$$\mu = \int_{-\infty}^{\infty} x\,f(x)\,dx$$

$$= \int_{-\infty}^{1} x\,f(x)\,dx + \int_1^3 x\,f(x)\,dx + \int_3^{\infty} x\,f(x)\,dx$$

$$= 0 + \int_1^3 x\,f(x)\,dx + 0$$

$$= \int_1^3 x\,f(x)\,dx$$

$$= \int_1^3 x\,\frac{1}{4}(x-1)^3\,dx$$

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$$= \frac{1}{4}\int_1^3 x\,(x-1)^3\,dx$$

$$= \frac{1}{4}\int_1^3 x\left(\left(x^3 - 1 + 1^3 - 3x^3 x\; 3^2 x 7\right) 3x\,dx\,(\cdot 1)\right)$$

$$= \frac{1}{4}\int_1^3 (x^4 - 3x^3 + 3x^2 - x)\,dx$$

$$= \frac{1}{4}\left[\frac{x^5}{5} - \frac{3x^4}{4} + \frac{3x^3}{3} - \frac{x^2}{2}\right]_1^3$$

$$= \frac{1}{4}\left[\left(\frac{3^5}{5} - \frac{3(3)^4}{4} + 3^3 - \frac{3^2}{2}\right) - \left(\frac{1^5}{5} - \frac{3(1)^4}{4} + 1 - \frac{1}{2}\right)\right]$$

$$= \frac{1}{4}\left[(48.6 - 60.75 + 27 - 4.5) - (0.2 - 0.75 + 1 - 0.5)\right]$$

$$= \frac{1}{4}\left[(10.35) - (-0.05)\right]$$

$$= \frac{1}{4}\left[10.4\right]$$

$$= 2.6$$

Q.No.

By solving ⑧ and ⑨, we get.

$A = 0.9905$ $\quad$ $B = 1.3002$

From ③, $Y = 0.9905 + 1.3002X$.

$\Rightarrow \log y = \log 0.9905 + \log e^{1.3002X}$

$\Rightarrow \log y = \log(0.9905e^{1.3002x})$

$\Rightarrow y = 0.9905e^{1.3002x}$.

$\therefore$ By using the method of least squares, the Constants, $a = 0.9905$, $b = 1.3002$.
The required equation is $y = 0.9905e^{1.3002x}$.

9b.)

Given,

x	65	66	67	67	68	69	70	72
y	67	68	65	68	72	72	69	71

Let the required equation be $y = ax + b$ -①

The equations be

$\Sigma y = a\Sigma x + nb$ -②

$\Sigma xy = a\Sigma x^2 + b\Sigma x$ -③

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x	y	x^2	xy
65	67	4225	4355
66	68	4356	4488
67	65	4489	4355
67	69	4489	4556
68	72	4624	4896
69	72	4761	4968
70	69	4900	4830
72	71	5184	5112

$\Sigma x = 544$ $\Sigma y = 552$ $\Sigma x^2 = 37,028$ $\Sigma xy = 37,560.$

By substituting these values in ② & ③, we get.

$552 = a\,544 + 8b$ —④

$37,560 = 37,028a + 544b$ —⑤

By solving these two regression lines we get.

$a = 0.666$ $b = 43.667.$

$\therefore$ $y = 0.666x + 3.667$ is the required line

The value of x when $y = 70$ is,

$70 = 0.666x + 3.667.$

$\Rightarrow$ $0.666x = 66.33$

$x = 99.44$

Q.No.

1a.)

Given,

there are three bags b_1, b_2, b_3 with equal chance,

$$\therefore P(b_1) = P(b_2) = P(b_3) = \frac{1}{3}.$$

In bag b_1, when two balls are drawn, Probability of one white and one red, is $P(E/b_1) = \frac{1}{6} \cdot \frac{2}{6} = \frac{2}{36} = \frac{1}{18}$.

In bag b_2, when two balls are drawn, Probability of one white and one red is $P(E/b_2) = \frac{2}{6} \cdot \frac{3}{6} = \frac{6}{36} = \frac{1}{6}$.

In bag b_3, when two balls are drawn, Probability of one white and one red is $P(E/b_3) = \frac{3}{6} \cdot \frac{1}{6} = \frac{3}{36} = \frac{1}{12}$.

$$\therefore P(b_1) = \frac{1}{3}, \quad P(E/b_1) = \frac{1}{18}.$$
$$P(b_2) = \frac{1}{3}, \quad P(E/b_2) = \frac{1}{6}.$$
$$P(b_3) = \frac{1}{3}, \quad P(E/b_3) = \frac{1}{12}.$$

By baye's theorem,
the Probability of the balls drawn from second bag is,

$$P(b_2/E) = \frac{P(b_2) \cdot P(E/b_2)}{P(b_1) \cdot P(E/b_1) + P(b_2) \cdot P(E/b_2) + P(b_3) \cdot P(E/b_3)}$$

$$= \frac{\frac{1}{3} \cdot \frac{1}{6}}{\frac{1}{3} \cdot \frac{1}{18} + \frac{1}{3} \cdot \frac{1}{6} + \frac{1}{3} \cdot \frac{1}{12}} = \frac{\frac{1}{18}}{\frac{1}{54} + \frac{1}{18} + \frac{1}{36}}$$

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$$P(b_2/E) = \frac{0.055}{0.0185 + 0.055 + 0.0277}$$

$$= \frac{0.055}{0.1012}$$

$$= 0.54347826.$$

∴ The Probability of that the balls drawn Came from bag b_2 is,

$$P(b_2/E) = 0.54347826.$$

1b)

Given,

a random variable X such that,

X	0	1	2	3	4	5	6	7
P(x)	0	K	2K	2K	3K	K^2	$2K^2$	$7K^2+K$

i) K

we know that $\Sigma P(x) = 1$

$\Rightarrow 0 + K + 2K + 2K + 3K + K^2 + 2K^2 + 7K^2 + K = 1$

$\Rightarrow 10K^2 + 9K = 1$

$\Rightarrow 10K^2 + 9K - 1 = 0$

$\Rightarrow 10K^2 + 10K - K - 1 = 0$

$\Rightarrow 10K(K+1) - 1(K+1) = 0$

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$\Rightarrow \quad K = 0.1$ (or) -1.

$\therefore \boxed{K = 0.1}$ as $K > 0$.

ii.) $P(X<6) = P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4)$
$+ P(X=5)$

(or)

$P(X<6) = 1 - P(X \geq 6)$
$= 1 - [P(X=6) + P(X=7)]$

$= 1 - [2(0.1)^2 + 7(0.1)^2 + (0.1)]$
$= 1 - [0.02 + 0.07 + 0.1]$
$= 1 - [0.19]$

$\therefore P(X<6) = 0.81$

iii.) $P(0<X<5)$

$= P(X=1) + P(X=2) + P(X=3) + P(X=4)$

$= K + 2K + 2K + 3K$

$= 0.1 + 0.2 + 0.2 + 0.3$

$\therefore P(0<X<5) = 0.8$

iv.) Mean, $\mu = \Sigma x p(x)$

$\mu = 0 \cdot 0 + 1 \cdot K + 2(2K) + 3(2K) + 4(3K) + 5(K^2) + 6(2K^2)$
$+ 7(7K^2 + K)$

$= 0 + K + 4K + 6K + 12K + 5K^2 + 12K^2 + 49K^2 + 7K$

$= 66K^2 + 30K$

$= 66(0.1)^2 + 30(0.1) = 3.66.$

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v.) Variance, $\sigma^2 = \Sigma x^2 p(x) - \mu^2$

$= 0(0) + 1(k) + 4(2k) + 9(2k) + 16(3k) + 25(k^2) + 36(2k^2) + 49(7k^2+k) - (3.66)^2$

$= k + 8k + 18k + 48k + 25k^2 + 72k^2 + 343k^2 + 49k - (3.66)^2$

$= 124k + 440k^2 - (3.66)^2$

$= 124(0.1) + 440(0.01) - (3.66)^2$

$= 12.4 + 4.4 - 13.3956$

$= 3.4044.$

$\therefore$ Variance, $\sigma^2 = 3.4044$

3 a) Given,

$n = 1000$ students.

mean $\mu = 78$

standard deviation $\sigma = 11$

i. above 90

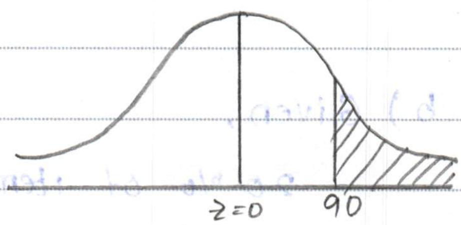
By normal distribution

$$z = \frac{x - \mu}{\sigma}$$

$$z = \frac{x - 78}{11}$$

i. above 90

$P(x > 90) = P(x = 90)$



$$z = \frac{90 - 78}{11}$$

$$= \frac{12}{11} = 1.09 > 0$$

$P(\cancel{x} > 90) = P(z > 90)$

$= 0.5 - A(1.09)$

$= 0.5 - 0.3621$

$= 0.1379$

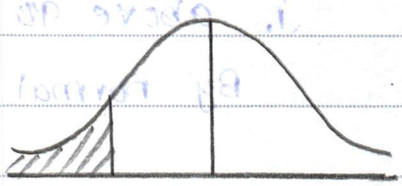
Q.No.

ii, below 60

$P(X<60) = P(X=60)$

$= \frac{60-78}{11} = \frac{-18}{11}$

$= -1.63 < 0$



$P(X<60) = P(Z<60)$

$= 0.5 - A(-1.63)$

$= 0.5 - A(1.63)$

$= 0.5 - 0.4484$

$= 0.0516.$

b) Given,

20 % of items produced from a factory are defective

let p be the probability of defective = 20 %

$= 0.20$

q be the probability of non-defective $= 1-p$

$= 1-0.20$

$= 0.8$

Since 5 items are chosen at random

$\therefore n = 5$

By binomial distribution

${}^{n}C_r \, p^r q^{n-r}$

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$${}^{5}C_{r}\ (0.2)^{r}\ (0.8)^{5-r}$$

i, none is defective

$$P(x=0) = {}^{5}C_{0}\ (0.2)^{0}\ (0.8)^{5}$$

$$= (1)(1).\ (0.32768)$$

$$= 0.32768$$

ii, One is defective

$$P(x=1)$$

$$= {}^{5}C_{1}\ (0.2)^{1}\ (0.8)^{5-1}$$

$$= 5\ (0.2)\ (0.8)^{4}$$

$$= (1)\ (0.4096)$$

$$= 0.4096$$

iii, at most 2 defectives

$$P(x \leq 2) = P(0) + P(1) + P(2)$$

$$= {}^{5}C_{0}\ (0.2)^{0}(0.8)^{5} + {}^{5}C_{1}\ (0.2)^{1}(0.8)^{4} +$$

$${}^{5}C_{2}\ (0.2)^{2}\ (0.8)^{3}$$

$$= 0.32768 + 0.4096 + 10 \times 0.04 \times 0.512$$

$$= 0.32768 + 0.4096 + 0.2048$$

$$= 0.8192.\ \ 0.94208$$

Q.No.

5 a) Data :

Data is a collection of facts, observations and measurements that are used for decision making, analysis and making pridiction. In statistics and data science data plays a very important role in understanding the problem, making pridictions and to solve a real world problems. Data is classified into various types and is collected in a systematic method.

Types of Data :- 1. Qualitative data
2. Quantitative data

Qualitative data :

Qualitative data refers to the data that describes qualities, characteristics and catagories. and these are non-numeric in nature. ex: gender, colours

characteristics:
* it cannot be measured numerically
* used to describe catogories.
* used for classification.

Types :- 1. Nominal
2. Ordinal.

i. Nominal :

Represents catogories in not a meaningful Order or Specific Order

eg: Religion, colour.

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ii, ordinal:

Represents categories in a meaningful order or in specific order.

ex: Feedback (good, better, best).

Quantitative data:

Quantitative data refers to the data that can be measured or counted. It is used in mathematical and statistical calculations.

ex: Height, weight, age.

Types: i. Discrete

ii. continuous.

i. Discrete:

- can take only whole numbers
- Obtained by counting

ex: No of Students

ii. Continuous:

- can take any number within the Range
- obtained by measurement

ex: Temperature, Humidity.

Collection of Data:

Data collection is a systematic method of gathering information from different sources. for Analysis, research and decision making. There are two major method of collection of data:

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1. primary data collection:

primary data collection is the data collected for the first time for the specific research purpose

characteristics.

- Fresh and Original
- accurate to current research requirements
- expensive

Sources of primary data collection:

- observation method
- experimental method
- Interview method
- Quationarre method
- Surveys.

2. Secondary data collection:

secondary data collection is the data collected for other purpose and is available for current research use

characteristics:

- less expensive
- may require cleaning
- come from different sources

Sources for secondary data collection:

- Books and Journals - surveys.
- Government publications
- Newspapers and Magzines
- Online database and wedsite

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b) Given,

numbers = 2, 3, 6, 8, 11

$n = 5$

i, mean of the population.

mean $\mu = \frac{\Sigma x_i}{n} = \frac{2+3+6+8+11}{5} = \frac{30}{5} = 6$

ii, standard deviation of population.

$S.D = \sigma^2 = \frac{\Sigma(x_i - \bar{x})^2}{n}$

$= \frac{(2-6)^2 + (3-6)^2 + (6-6)^2 + (8-6)^2 + (11-6)^2}{6}$

$= \frac{16 + 9 + 0 + 4 + 25}{6} = \frac{54}{6} = 9$

$= 9$

iii, mean of sampling distibution.

drawn with replacement $= N^n$ (∴ given $n=2$)

$= 5^2 = 25$

possible outcomes: (2,3) (2,6) (2,11) (2,8) (2,2) (8,2)

(3,3) (3,2) (3,6) (3,8) (3,11) (8,3)

(2,6) (6,3) (6,6) (6,8) (6,11) (8,6)

(8,8) (11,2) (11,6) (11,11) (11,3) (11,8) (8,11)

mean of each sample = 2.5, 4, 6.5, 5, 2, 8, 7,

3, 2.5, 4.5, 5.5, 7, 5,

4, 4.5, 6, 7, 8.5, 5.5,

6.5, 8.5, 11, 7, 9.5, 9.5

mean $\bar{x} = \frac{2.5+4+6.5+5+2+3+2.5+4.5+5.5+7+4+4.5+6+7+8.5+6.5+8.5+11+7+9.5+8+7+5+5.5+9.5}{25}$

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Mean $\bar{x} = \frac{150}{25} = 6$

iv, standard deviation of sampling distribution of means.

$S.D = \sigma^2 = \frac{\Sigma(x_i - \bar{x})^2}{N}$

$= (2.5-6)^2 +$

$= \frac{3.5^2 + 2^2 + 0.5^2 + 1^2 + 4^2 + 3^2 + 3.5^2 + 1.5^2 + 0.5^2 + 1^2 + 2^2 + 1.5^2 + 1^2 + 2.5^2 + 0.5^2 + 2.5^2 + 5^2 + 1^2 + 3.5^2 + 2^2 + 1^2 + 1^2 + 1.5^2 + 3.5^2}{25}$

$= \frac{137}{25} = 5.48$

$\sigma = \sqrt{5.48} = 2.34$

8) Given that,

two Factories produces electric bulbs.

	Factory A	Factory B
No. of bulbs	100	200
Average life	1,100 hours	900 hours
Standard deviation	240 hours	220 hours.

$\therefore n_1 = 100$, $n_2 = 200$

$\bar{x}_1 = 1,100$ $\quad \bar{x}_2 = 900$

$\sigma_1 = 240$ $\quad \sigma_2 = 220$

Q.No.

$\Sigma XY = 24$, $\Sigma X^2 = 36$, $\Sigma Y^2 = 44$

two regression lines are

x on y

$x - \bar{x} = bxy\,(y - \bar{y})$ — (1)

y on x

$y - \bar{y} = byx\,(x - \bar{x})$ — (2)

from (1)

$x - 68 = bxy\,(y - 69)$

$bxy = \frac{\Sigma XY}{\Sigma Y^2} = \frac{24}{44} = 0.54$

$x - 68 = 0.54\,(y - 69)$

$x - 68 = 0.54y - 37.26 \Rightarrow x = 0.54y + 30.74$

when $y = 70$

$x - 68 = 0.54(70) - 37.26$

$x - 68 = 37.8 - 37.26$

$x = 0.54 + 68$

$x = 68.54$

from (2)

$y - 69 = byx\,(x - 68)$

$byx = \frac{\Sigma XY}{\Sigma X^2} = \frac{24}{36} = 0.6666 \simeq 0.67$

$y - 69 = 0.67\,(x - 68)$

$y - 69 = 0.67x - 45.56$

$y = 0.67x + 23.44$

Q.No.

when $y = 70$

$$70 = 0.67x + 23.44$$

$$\Rightarrow x = \frac{46.56}{0.67}$$

$$x = 69.49.$$

$\therefore$ The two regression lines are

$x = 0.54y + 30.74$ when $y = 70, x = 68.54$

and $y = 0.67x + 23.44$ when $y = 70, x = 69.49.$

7 a) Given,

$n = 49$

$\mu = 15200$

Sample mean $\bar{x} = 15150$

Standard Deviation $= 120$

Null Hypothesis : $H_0 : \mu = 15200$

Alternative Hypothesis $H_1 : \mu \cancel{\times} 15200$

~~Crit~~ level of significance $= 0.05$

critical value $z_\alpha = 1.96$

Test statistic :

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$$= \frac{15150 - 15200}{\frac{120}{\sqrt{49}}}$$

Q.No.

$$z = \frac{-\cdot 50}{\frac{120}{7}}$$

$$= \frac{-35}{12}$$

$$= -2.91 < 0.$$

$$|z_{cal}| = |-2.91|$$
$$= 2.91$$

$\therefore |z_{cal}| > |z_{tab}|$

$\therefore$ calculated value is greater than tabulated value

$\therefore$ H_0 is rejected. (null hypothesis is not accepted).
H_1 is accepted.

b) Given

$n_1 = 6400$ $n_2 = 1600$

$\bar{x}_1 = 67.85$ $\bar{x}_2 = 68.55$

$\sigma_1 = 2.56$ $\sigma_2 = 2.52$

Null Hypothesis : $H_0 : \mu_1 = \mu_2$

Alternative Hypothesis : $H_1 : \mu_1 \neq \mu_2$

level of significance $\alpha = 0.01$

critical value $z_\alpha = 2.33$

Test statistic :

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Q.No.

$$z = \frac{67.85 - 68.55}{\sqrt{\frac{(2.56)^2}{6400} + \frac{(2.52)^2}{1600}}}$$

$$= \frac{-0.7}{\sqrt{\frac{6.5536}{6400} + \frac{6.3504}{1600}}}$$

$$= \frac{-0.7}{0.070661}$$

$$= -9.906$$

$|z_{cal}| = |-9.906|$

$= 9.9$

$\simeq 10$

$\therefore |z_{cal}| > z_{tab}$

$\therefore$ calculated valued of z is greater than tabulated value of z

$\therefore$ H_0 is not accepted.

$\therefore$ Australians are not taller than Englishmen.

Q.No.

6a) Given

$n = 100$

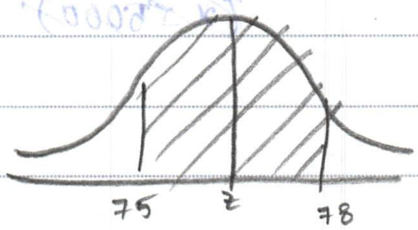
$\mu = 76$

$\sigma^2 = 256 \Rightarrow \sigma = 16$

$P(75 < \bar{x} < 78)$

By central Limit theorem

$$z = \frac{x - \mu}{\sigma}$$



when $x = 75 \Rightarrow z_1 = \frac{75 - 76}{16}$

$= \frac{-1}{16} = -0.06 < 0$

when $x = 78$

$$z_2 = \frac{78 - 76}{16}$$

$= \frac{2}{16} = 0.125 > 0$

$P(75 < \bar{x} < 78) = P(75 < z < 78)$

$= P(z_1 < z < z_2)$

$= A(z_1) + A(z_2)$

$= A(-0.06) + A(0.125)$

$= 0.0239 + 0.0478$

$= 0.0718$

Q.No.

b) Given

$\mu = 1500$, $n = 3$

$S.D = 150$

i, at least 5,000

$$z = \frac{x - \mu}{\sigma}$$

~~(x ≤ 5000) = 1 - p(x>~~

Q.No.

Q.No.

Q.No.

Q.No.